

Introduction to Generative AI

Generative Artificial Intelligence, commonly known as Generative AI, is a type of artificial intelligence that can create new content such as text, images, audio, video, and computer code.

Unlike traditional AI systems that mainly classify data or make predictions, Generative AI learns patterns from large amounts of training data and uses those patterns to generate new content.

Large Language Models

Large Language Models, also known as LLMs, are AI models trained on very large amounts of text data. They can understand and generate human-like text.

Examples of popular Large Language Models include GPT, Gemini, Claude, Llama, and Mistral.

LLMs can perform many tasks such as:

- Question answering
- Text summarization
- Translation
- Content generation
- Code generation
- Chatbots

LangChain

LangChain is an open-source framework used to build applications powered by Large Language Models.

It helps developers connect LLMs with different components such as prompts, document loaders, text splitters, embeddings, vector databases, retrievers, tools, and external data sources.

For example, LangChain can be used to build an AI application where a student uploads a PDF and asks questions about the information contained in that PDF.

Embeddings

Embeddings are numerical vector representations of data that capture semantic meaning.

An embedding model converts text into a list of numbers called a vector.

For example:

"Artificial Intelligence"

can be converted into a vector such as:

[0.21, -0.45, 0.78, 0.32, ...]

Texts with similar meanings generally have embeddings that are closer to each other in vector space.

Embeddings are commonly used for semantic search, recommendation systems, similarity search, and Retrieval-Augmented Generation.

Vector Databases

A vector database stores and searches vector embeddings.

In a RAG application, document chunks are converted into embeddings and stored in a vector database along with their original text and metadata.

Examples of vector databases and vector stores include FAISS, Chroma, Pinecone, and Weaviate.

Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, is a technique that combines information retrieval with Large Language Models.

A basic RAG system works in the following steps:

1. Load documents.
2. Split documents into smaller chunks.
3. Convert the chunks into embeddings.
4. Store the embeddings in a vector database.
5. Accept a question from the user.
6. Convert the question into an embedding.
7. Search for relevant document chunks.
8. Retrieve the most relevant chunks.
9. Provide those chunks as context to an LLM.
10. Generate the final answer.

CourseMate AI

CourseMate AI is an AI-powered study assistant that allows students to interact with their learning materials.

Students can upload resources such as lecture notes, textbooks, PDFs, and research papers.

Instead of manually searching through long documents, students can ask questions in natural language.

For example, a student may ask:

"What is Retrieval-Augmented Generation?"

CourseMate AI searches the uploaded study material, retrieves the most relevant information, and provides that information to an LLM.

The LLM then generates an easy-to-understand answer based on the retrieved context.

The main goal of CourseMate AI is to make studying faster and easier by allowing students to chat with their own study materials.